

Operate a Press Brake Mock SOP (1x Episode: one sheet, three bends)

One episode bends **one** sheet on the press brake mock. Put the sheet against the back stops. Cycle the press. Turn the part. Do this three times, for three bends. Then gauge the three angles. Put the part in the finished tray. The episode ends when the part lies in the finished tray or in the reject bin, the gauge is back on the gauge mat, and the press is up. Never split one sheet into two episodes.

The base is **passive**. It has no motor. You push it by hand to the bench in front of the press brake mock and lock it there. Nothing is carried away to another table. Everything the episode touches is already there when recording starts: the press brake mock bolted to the bench, the blank tray with flat sheets in it, the finished tray, the reject bin, and the angle gauge on the gauge mat.

The press is used **as found**. It starts up, with nothing in it. The sheet is a flat blank with three bend lines drawn across it. Each line has a number on it: 1, 2, and 3. The episode ends with the press up and empty, ready for the next sheet.

This is a bench task. Four rules come from that. First, the **press brake mock stays bolted down**. You never push, pull, lift, or lean on the press body or the bench. Second, **the lever only goes down and up**. You pull it down until it stops, and you lift it back up until it stops. You never twist it, jerk it, or let it drop. Third, **nothing is in the gap while the press moves, except the sheet**. No gripper, wrist, or tool ever goes under the punch. You hold the sheet by its front edge, in front of the die, and nowhere else. Fourth, **the sheet is bent on its line, and nowhere else**. The line sits over the die groove before the press comes down. If the line is off, you fix it before you cycle. You never cycle to see what happens.

A bend cannot be undone. A part bent off its line, or bent on the wrong line, or bent the wrong way up, is a reject. You do not bend it again. You do not try to flatten it. You put it in the reject bin and report it.

The order never changes: **sheet against the stops; cycle; turn; sheet against the stops; cycle; turn; sheet against the stops; cycle; then gauge all three angles; then tray or bin.**

Each gripper holds one thing at a time. Nothing is passed from one gripper to the other. The sheet is never in the air while the lever moves.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the bench. Stop it **square to the front edge of the bench**, so the front edge runs straight across the frame and both ends look the same distance away.
2. Stop it **in the middle of the press brake mock**, so the blank tray side and the finished tray side are the same distance from the middle of the base.
3. Stop it **close enough** that a gripper holding the sheet by its front edge can push the sheet back until it touches both back stops, without stretching the arm. Stop it **far enough** that no arm, wrist, or part of the base touches the bench front, the press body, or the lever while both arms work.
4. Check the **lever**. With the base parked, a gripper closes on the lever knob and pulls the lever all the way down and lifts it all the way up without stretching.
5. Check the **stops**. A gripper holding a sheet by its front edge pushes the sheet back until it touches both back stops and holds it there without stretching. The gripper stays in front of the die the whole time.
6. Check the **bench**. A gripper reaches the top sheet in the blank tray, the gauge on the gauge mat, the finished tray, and the reject bin. It does all four without stretching and without knocking anything over.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
8. If any line from 1 to 6 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bench the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bench hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the press brake mock. The frame shows the whole press, the punch, the die, both back stops, the lever, the blank tray, the gauge mat with the gauge, the finished tray, and the reject bin.
3. The camera sees the die from the front, so you can see the sheet lying on the die and see the bend line over the die groove.
4. The camera sees the back stops, so you can see if the sheet touches both stops or only one.

5. The camera sees the punch and the lever, so you can see if the press is up or down.
6. The camera sees the gauge mat from above, so you can see the part on the mat and the gauge held in each bend.
7. The camera sees the finished tray pockets, so you can see where the part lands.
8. Both arms are at home with the grippers open.
9. The grippers reach the top sheet in the blank tray, the front edge of a sheet on the die, the lever knob, the gauge, the finished tray, and the reject bin without stretching to their limit.
10. The two arms do not hit each other when one holds the sheet on the die and the other works the lever.
11. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 9 and 10 are all true.

Materials checklist

1. The **press brake mock** is a small hand press made for practice. It bends thin sheet with a lever. It has no motor and no foot pedal. Nothing is powered.
2. The **press body** is bolted to the bench. Nobody moves it, leans on it, or stands anything on it.
3. The **die** is the bottom block. It has one groove along its top, running left to right. The sheet lies on the die with its bend line over the groove.
4. The **punch** is the top blade. It comes straight down into the groove when the lever is pulled, and goes straight up when the lever is lifted. It stops hard at the top and at the bottom.
5. The **gap** is the space between the punch and the die. Only the sheet is ever in it.
6. The **back stops** are two fingers behind the die. They stand at the same distance from the groove. A sheet pushed back until it touches both fingers has its line over the groove. Nobody moves the stops during the episode.
7. The **lever** is a bar with a **knob** on its end, on the side of the press. Pull the knob down until it stops to bend. Lift the knob up until it stops to open the press.
8. The lever moves easily by hand. Test it by hand during setup. If it sticks, stop and fix the press. Nobody forces it during the episode.
9. The **sheet** is one flat blank of thin, soft sheet. It has a **top face** with three bend lines drawn across it. Each line has a number next to it: **1**, **2**, and **3**. Line 1 is near one end. Line 2 is near the other end. Line 3 is between line 1 and the middle.
10. The sheet bends at **90 degrees** on each line. The press is set to 90 degrees and stays there. Nobody changes the press setting during the episode.
11. The finished part is a **C**: a flat bottom, a wall at each end, and a short lip on top of one wall that points in.
12. The **blank tray** is a flat tray on one side of the press. Flat sheets lie in it in a stack, top face up, with line 1 toward the back. The top sheet is the one you take.

13. The **gauge** is a small metal square. Its two edges meet at 90 degrees. You set it into a bend to check the angle.
14. The **gauge mat** is a rubber mat on the bench, in front of the blank tray. The gauge lives on it. The part is laid on it to be gauged. Nothing else is on it.
15. The **finished tray** is a flat tray with soft pockets on the other side of the press. Each pocket holds one part, bottom down. It has at least one empty pocket.
16. The **reject bin** is an open bin behind the finished tray. A part with a bad bend goes in it.
17. The bench top is clean and clear at the start. No sheet, part, or scrap lies on it, on the die, or on the gauge mat.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bench and the press themselves.

- **Press brake mock:** the middle of the bench. Die at the front, back stops behind it, punch above, lever on the side. Never moved.
- **Die and groove:** where the sheet lies for every bend. The bend line sits over the groove.
- **Back stops:** the two fingers behind the die. The sheet is pushed back until it touches both.
- **Front edge zone:** the strip of sheet that sticks out in front of the die. This is the only place a gripper ever holds the sheet while it is on the die.
- **Lever and knob:** on the side of the press. One gripper on the knob, and only on the knob.
- **Blank tray:** on one side of the press. Flat sheets start here. Only the top sheet is taken.
- **Gauge mat:** in front of the blank tray. The gauge starts and ends here. The part is laid here to be gauged.
- **Finished tray:** on the other side of the press. Good parts go here, one to a pocket, bottom down.
- **Reject bin:** behind the finished tray. A part with a bad bend goes here.

Gripper rules

- **Sheet:** one gripper on the sheet, by its front edge, in the middle of that edge. That gripper stays in front of the die and never goes under the punch.
- **Lever:** one gripper on the knob, and only on the knob. It never touches the lever bar, the punch, or the press body.
- **Gauge:** one gripper on the gauge, by its outside corner. The gauge is never in a gripper while the lever moves.
- **Hold:** the gripper on the sheet keeps the sheet pushed against both back stops from before the press starts down until after the press is all the way up. A hold never pulls or turns the sheet while the press moves.
- Nothing is passed from one gripper to the other. Only one thing moves at a time.

Vocabulary

- **Press up:** the punch is at its top stop. The gap is open. The lever is all the way up.
- **Press down:** the punch is at its bottom stop, in the groove. The lever is all the way down.
- **Cycle:** pull the lever down until it stops, hold it there for 1 second, then lift it up until it stops. One cycle makes one bend.
- **Against the stops:** the back edge of the sheet touches both back stops at the same time. Not one. Both.
- **On the line:** the numbered bend line lies right over the die groove, straight, from side to side.
- **Front edge grip:** the gripper closes on the sheet in the middle of its front edge, top and bottom, flat, in front of the die.
- **Half turn:** the sheet stays flat on the bench and is turned around, end for end, so the end that was at the front is now at the back. Top face stays up. The sheet is never flipped over.
- **Top face up:** the face with the numbers on it points at the ceiling.
- **Low carry:** the sheet or the part is carried close over the bench and level. It never goes over the press.
- **Wall:** the part of the sheet that stands up after a bend.
- **Lip:** the short wall made by bend 3. It points in, over the bottom of the part.
- **Gauged:** the gauge is set into the inside of a bend, with one edge flat on the bottom and the other edge flat on the wall, and held still for 2 seconds in front of the camera.
- **Good bend:** the gauge sits flat on both faces with no gap showing on either edge.
- **Bad bend:** a gap shows between the gauge and the bottom or between the gauge and the wall. Or the bend is not on its line. Or the bend is the wrong way up.
- **Seated in the pocket:** the part sits bottom down in one finished tray pocket, inside the pocket walls, and does not rock.
- **Reset:** the press is up and empty, the gauge lies on the gauge mat, and both grippers are open and clear.
- **Back and clear:** the arm is pulled back, so no part of it is over the press, the gauge mat, or the trays.

Steps

Step 1: Take a sheet to the press

Goal: the flat sheet lies on the die, top face up, with line 1 at the back.

1.1 Take the top sheet

- Close a gripper on the top sheet in the blank tray, in the middle of its front edge, in a **front edge grip**.

- Lift it just off the sheet under it. Do not lift two sheets.

Check: one sheet hangs flat in the gripper, top face up, with the numbers showing. If two sheets came up, put them back and take one.

1.2 Lay it on the die

- Carry the sheet to the press in a **low carry**. Do not carry it over the press.
- Slide it in from the front, flat, over the die, with line 1 toward the back stops.
- Keep the gripper closed on the front edge. Keep the gripper in front of the die.

Check: the sheet lies flat on the die, **top face up**, with line 1 at the back. The gripper is in front of the die, not under the punch. If the sheet is upside down, take it back to the blank tray, turn it over there, and take it again.

Expected state: the sheet lies flat on the die with line 1 toward the back. One gripper holds its front edge. The press is up.

Step 2: Bend 1

Goal: the sheet is bent at 90 degrees on line 1, and the press is up.

2.1 Push the sheet against the stops

- Push the sheet straight back, flat on the die, until its back edge touches both back stops.
- Push gently. Do not lift the sheet. Do not turn it.
- Keep the gripper closed and keep pushing lightly, so the sheet stays against the stops.

Check: the sheet is **against the stops**, both of them. Line 1 is **on the line**, over the groove, straight from side to side. If only one stop is touched, ease off and push again, square. If the line is off the groove, stop and report it. Do not cycle.

2.2 Cycle the press

- With the other gripper, close on the lever knob.
- Pull the lever down, slow and smooth, until it stops. Hold it there for 1 second.
- Lift the lever up, slow and smooth, until it stops.
- Keep the **hold** on the sheet the whole time. The sheet does not move until the press is up.

Check: the **press is up**. The sheet has a wall standing up at the back, on line 1. The gripper on the sheet never went under the punch. If the lever stopped short of the bottom, pull it down again until it stops, then lift it up.

2.3 Let go of the lever

- Open the gripper on the knob and pull it back from the lever.

Expected state: the sheet has one wall, at the back. The press is up. One gripper still holds the front edge. The other gripper is back and clear.

Step 3: Turn the part and bend 2

Goal: the sheet is bent at 90 degrees on line 2, and the press is up.

3.1 Take the part out

- Pull the part straight forward, flat, out of the press, until the wall is clear of the punch.
- Lay it flat on the bench in front of the press. Open the gripper.

3.2 Half turn

- Close on the part again by the back edge, now free. Turn it flat on the bench, a **half turn**, end for end.
- Do not flip it over. The top face stays up.

Check: line 2 is now at the back, and wall 1 is at the front, pointing up. The numbers still face up. If the part is upside down, turn it back over and turn it again, flat.

3.3 Lay it on the die

- Close on the part in a **front edge grip**, on the flat sheet just in front of wall 1. Do not close on the wall.
- Slide it in from the front, flat, over the die, with line 2 toward the back stops.
- Keep the gripper in front of the die.

3.4 Push against the stops

- Push the part straight back, flat, until its back edge touches both back stops.
- Keep the gripper closed and keep pushing lightly.

Check: the part is **against the stops**, both of them. Line 2 is **on the line**, over the groove. If only one stop is touched, ease off and push again, square. If the line is off the groove, stop and report it. Do not cycle.

3.5 Cycle the press

- With the other gripper, close on the lever knob.
- Pull the lever down, slow and smooth, until it stops. Hold it there for 1 second.
- Lift the lever up, slow and smooth, until it stops.
- Keep the **hold** on the part the whole time.

Check: the **press is up**. The part now has two walls, one at each end, both pointing up. If the lever stopped short of the bottom, pull it down again until it stops, then lift it up.

3.6 Let go of the lever

- Open the gripper on the knob and pull it back from the lever.

Expected state: the part has two walls. The press is up. One gripper still holds the front edge. The other gripper is back and clear.

Step 4: Turn the part and bend 3

Goal: the sheet is bent at 90 degrees on line 3, and the press is up.

4.1 Take the part out

- Pull the part straight forward, flat, out of the press, until wall 2 is clear of the punch.
- Lay it flat on the bench in front of the press. Open the gripper.

4.2 Half turn

- Close on the part again and turn it flat on the bench, a **half turn**, end for end.
- Do not flip it over. The top face stays up.

Check: wall 1 is now at the back and wall 2 is at the front. Line 3 lies between wall 1 and the middle. If the part is upside down, turn it back over and turn it again, flat.

4.3 Lay it on the die

- Close on the part in a **front edge grip**, on the flat sheet just in front of wall 2. Do not close on the wall.
- Slide it in from the front, flat, over the die, with wall 1 toward the back stops.
- Keep the gripper in front of the die.

4.4 Push against the stops

- Push the part straight back, flat, until wall 1 touches both back stops.
- Keep the gripper closed and keep pushing lightly.

Check: the part is **against the stops**, both of them. Line 3 is **on the line**, over the groove. If only one stop is touched, ease off and push again, square. If the line is off the groove, stop and report it. Do not cycle.

4.5 Cycle the press

- With the other gripper, close on the lever knob.
- Pull the lever down, slow and smooth, until it stops. Hold it there for 1 second.
- Lift the lever up, slow and smooth, until it stops.
- Keep the **hold** on the part the whole time.

Check: the **press is up**. The part is now a **C**: a flat bottom, two walls, and a lip on top of wall 1 that points in. If the lever stopped short of the bottom, pull it down again until it stops, then lift it up.

4.6 Let go of the lever and take the part out

- Open the gripper on the knob and pull it back from the lever.
- Pull the part straight forward, flat, out of the press, until the lip is clear of the punch.

Expected state: the part is a C with three bends. The press is up and empty. One gripper holds the part by its front edge. The other gripper is back and clear.

Step 5: Gauge the three angles

Goal: you have seen the gauge in each of the three bends in front of the camera, and you know if each bend is good or bad.

5.1 Lay the part on the gauge mat

- Carry the part to the gauge mat in a **low carry**. Do not carry it over the press.
- Lay it down on the mat, bottom down, with the open side toward the camera.
- Open the gripper and pull it back.

Check: the part sits flat on the mat, bottom down, and does not rock. If it rocks, take it by its front edge, lift it, and put it down again.

5.2 Pick up the gauge

- Close a gripper on the gauge by its outside corner.
- Lift it just off the mat.

5.3 Gauge bend 1

- Set the gauge into the inside of bend 1. One edge flat on the bottom. The other edge flat on wall 1.
- Hold it still for **2 seconds** in front of the camera.

Check: the bend is **gauged**. You can see both gauge edges against the part. If the gauge is on the outside of the bend, lift it and set it on the inside.

5.4 Gauge bend 2

- Lift the gauge and set it into the inside of bend 2. One edge flat on the bottom. The other edge flat on wall 2.
- Hold it still for **2 seconds** in front of the camera.

5.5 Gauge bend 3

- Lift the gauge and set it into the inside of bend 3. One edge flat on wall 1. The other edge flat on the lip.
- Hold it still for **2 seconds** in front of the camera.

5.6 Put the gauge back

- Lift the gauge off the part and lay it flat on the gauge mat where it started, away from the part.
- Open the gripper and pull it back.

5.7 Decide good or bad

- If all three bends are **good bends**, the part is good. Go to Step 6.
- If any bend is a **bad bend**, the part is a reject. Close on the part by its front edge, carry it in a low carry to the reject bin, open the gripper over the bin, and let it drop in. Then go to Step 7.
- Do not bend a bad part again. Do not push a bad bend by hand to fix it.

Expected state: the part is either on the gauge mat, ready for the tray, or in the reject bin. The gauge lies on the mat.

Step 6: Put the part in the finished tray

Goal: the part sits bottom down in one empty tray pocket.

6.1 Carry it to the tray

- Close on the part in a **front edge grip**. Lift it just off the mat.
- Carry it in a **low carry** to the finished tray. Do not carry it over the press.
- Bring it over the nearest empty pocket.

6.2 Lower it into the pocket

- Lower the part straight down into the pocket, bottom down.
- Open the gripper and pull it back.

Check: the part is **seated in the pocket**. It sits inside the pocket walls, bottom down, and does not rock. If it sits on the pocket wall, or on its side, take it by its front edge, lift it, and put it down again.

Expected state: the part sits in a tray pocket. The gauge lies on the mat. The press is up and empty.

Step 7: Check the press and end the episode

- Check the press: it is up, and the gap is empty.
- Check the back stops: they stand where they started.
- Check the part: it sits bottom down in one finished tray pocket, or it is in the reject bin.
- Check the gauge: it lies flat on the gauge mat.
- Check the bench: no sheet or part lies on it, in front of the press, or on the die.
- Check the base has not moved.
- Fix any failed check before you end.
- Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Check the press is up by hand. Lift the lever to its top stop if it is not.
2. Check the gap is empty. Take out any part or scrap by hand.
3. Check the back stops: both stand at the same distance from the groove and do not wobble. Set them again if one moved.
4. Check the blank tray has at least one flat sheet in it, top face up, with line 1 toward the back. Add sheets if it is empty.
5. Check the top sheet: its three lines are clear and numbered, and it lies flat. Change it if it is bent, marked wrong, or has no numbers.
6. Check the finished tray has at least one empty pocket. Change the tray if it is full.
7. Empty the reject bin if it is more than half full.
8. Check the gauge: its edges are clean and straight. Lay it flat on the gauge mat, away from the front edge of the mat.
9. Check the lever: it goes all the way down and all the way up by hand with no catch.
10. Check the press body is still bolted tight and does not rock.
11. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the **start timestamp**, the **violation name**, and the **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the bench moves in the frame, the press changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.
- **SOP rule broken:** Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Gripper under the punch

- **Visible cue:** a gripper, wrist, or forearm goes into the gap between the punch and the die at any time. Or the gripper holds the sheet so close to the die that it is under the punch when the lever moves.
- **SOP rule broken:** Steps 2.2, 3.5, and 4.5. Only the sheet is ever in the gap. The gripper holds the front edge, in front of the die.
- **Coaching note:** front edge only. If the punch could touch the gripper, the gripper is in the wrong place.

Violation: Cycled without a hold

- **Visible cue:** the lever goes down while no gripper holds the sheet. Or the gripper lets go of the sheet before the press is all the way up. Or the sheet is pulled or turned while the punch is down.
- **SOP rule broken:** Steps 2.2, 3.5, and 4.5. Keep the hold on the sheet from before the press starts down until after it is all the way up.
- **Coaching note:** hold first, cycle second, hold until it is up. A sheet that moves under the punch is a reject.

Violation: Cycled off the stops

- **Visible cue:** the lever goes down while the sheet touches only one back stop, or neither. Or the sheet is not straight on the die. Or the bend line is clearly off the groove.
- **SOP rule broken:** Steps 2.1, 3.4, and 4.4. Push the sheet against both stops, check the line is on the groove, and only then cycle.
- **Coaching note:** both stops, then look at the line, then cycle. You never cycle to see what happens.

Violation: Bent on the wrong line

- **Visible cue:** bend 1 is made on line 2 or line 3. Or a bend is made where there is no line. Or a line is bent twice.
- **SOP rule broken:** Steps 2, 3, and 4. Line 1, then line 2, then line 3. One bend per line.
- **Coaching note:** read the number before you push it back. The stops set the place, but you set the line.

Violation: Wrong turn

- **Visible cue:** the part is flipped over between bends, so the numbers face down. Or it is turned a quarter turn instead of a half turn. Or it goes back in the press with no turn at all.
- **SOP rule broken:** Steps 3.2 and 4.2. Between bends the part gets a half turn, flat, end for end. It is never flipped over.
- **Coaching note:** end for end, numbers up. Two half turns, and nothing else.

Violation: Lever forced or dropped

- **Visible cue:** the lever is jerked, twisted, pulled sideways, or let go so it drops on its own. Or it is pushed past its stop.
- **SOP rule broken:** Steps 2.2, 3.5, and 4.5. Pull the lever down slow and smooth until it stops, hold for 1 second, then lift it up slow and smooth until it stops.
- **Coaching note:** slow down, hold, slow up. The stops do the stopping, not the arm.

Violation: Press not cycled all the way

- **Visible cue:** the lever stops short of the bottom, so the bend is not finished. Or the lever is left part way up, so the press is not up when the sheet is pulled out.
- **SOP rule broken:** Steps 2.2, 3.5, and 4.5. Down until it stops, then up until it stops.
- **Coaching note:** to the stop both ways. Almost down is not bent, and almost up is not open.

Violation: Press body touched or leaned on

- **Visible cue:** a gripper, wrist, or forearm rests on the press body, the punch, the back stops, or the bench. Or a back stop is pushed or knocked out of place.
- **SOP rule broken:** Steps 1 to 7. Nobody pushes, pulls, lifts, or leans on the press body or the bench. Nobody moves the stops.
- **Coaching note:** the arm holds itself up. If something moves when you touch it, you touched it too hard.

Violation: Held by the wrong part

- **Visible cue:** the gripper closes on the sheet by a side edge, by a wall, or by the lip instead of the front edge. Or it closes on the lever bar instead of the knob. Or it closes on the gauge by its edge instead of its corner.
- **SOP rule broken:** Steps 1.1, 2.2, 3.3, 4.3, and 5.2. Front edge on the sheet, knob on the lever, corner on the gauge.
- **Coaching note:** front edge, knob, corner. Three things, three grips.

Violation: Two sheets taken

- **Visible cue:** two sheets come up out of the blank tray together, or two sheets go onto the die.
- **SOP rule broken:** Step 1.1. Take the top sheet and only the top sheet.
- **Coaching note:** one sheet, one episode. If two come up, put them both back and start again.

Violation: Sheet laid down wrong

- **Visible cue:** the sheet goes onto the die upside down, with the numbers face down. Or with line 2 or line 3 at the back on the first bend. Or on a slant across the die.
- **SOP rule broken:** Step 1.2. Lay the sheet flat, top face up, with line 1 toward the back stops.

- **Coaching note:** numbers up, 1 to the back. If you cannot read the number, you cannot bend the line.

Violation: Carried over the press

- **Visible cue:** the sheet, the part, or the gauge crosses over the press. Or a thing is carried high instead of low and level.
- **SOP rule broken:** Steps 1.2, 5.1, 5.7, and 6.1. Every carry is low and level and never crosses the press.
- **Coaching note:** low and around. Nothing crosses the press.

Violation: Bend not gauged

- **Visible cue:** the part goes from the press to the tray with no stop at the gauge mat. Or one of the three bends is skipped. Or the gauge is set on the outside of a bend, or held for less than 2 seconds. Or the gauge is turned away from the camera.
- **SOP rule broken:** Steps 5.3, 5.4, and 5.5. Set the gauge into the inside of each bend and hold it still for 2 seconds in front of the camera.
- **Coaching note:** three bends, three gauges, two seconds each. If the camera did not see the gauge, nobody checked the angle.

Violation: Wrong call on the part

- **Visible cue:** a part with a gap at the gauge, or a bend off its line, goes into the finished tray. Or a part with three good bends goes into the reject bin.
- **SOP rule broken:** Step 5.7. Three good bends to the tray. Any bad bend to the reject bin.
- **Coaching note:** no gap, in the tray. Any gap, in the bin. There is no in between.

Violation: Bad part bent again

- **Visible cue:** a part goes back into the press after the gauge. Or a gripper pushes on a wall or the lip to change the angle by hand.
- **SOP rule broken:** Step 5.7. Do not bend a bad part again. Do not push a bad bend to fix it.
- **Coaching note:** a bend cannot be undone. A bad bend is a reject, not a retry.

Violation: Gauge put back wrong

- **Visible cue:** the gauge is left on the part, on the bench, off the mat, or in the press. Or it is dropped onto the mat.
- **SOP rule broken:** Step 5.6. Lay the gauge flat on the gauge mat where it started.
- **Coaching note:** back where it was. The next episode starts from there.

Violation: Part not seated in the pocket

- **Visible cue:** the part ends up on a pocket wall, across two pockets, on its side, rocking, or on top of another part.

- **SOP rule broken:** Step 6.2. Lower the part straight down into one empty pocket, bottom down, so it sits flat.
- **Coaching note:** straight down into one pocket, bottom down. Watch it settle before you let go.

Violation: Press left with something in it

- **Visible cue:** the episode ends with the part, a sheet, or scrap still on the die or in the gap.
- **SOP rule broken:** Step 7. The press ends up and empty.
- **Coaching note:** look in the gap before you go home. Empty means nothing on the die.

Violation: Done in the wrong order

- **Visible cue:** the lever goes down before the sheet is against the stops. Or the part goes to the tray before it is gauged. Or bend 2 is made before bend 1, or bend 3 before bend 2. Or the gauge comes out before the third bend.
- **SOP rule broken:** Steps 1 to 6. Stops, cycle, turn, three times. Then gauge. Then tray or bin. One thing at a time.
- **Coaching note:** the order is the task. Skipping a step is not faster, it is a lost episode.

Violation: Something dropped or knocked over

- **Visible cue:** the sheet, the part, or the gauge slips out of a gripper and falls on the bench or the floor. Or the blank tray, the finished tray, or the reject bin is knocked over by an arm going past.
- **SOP rule broken:** Steps 1 to 6. One thing moves at a time, and each one is put down under control.
- **Coaching note:** close all the way before you lift, and put it down before you let go.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails, and the episode carries on with no retry. For example, the sheet touches one stop, the lever stops short, the gauge is on the outside of a bend, or the part rocks in the pocket.
- **SOP rule broken:** Steps 1 to 6. Do the retry written under each failed check before you move on.
- **Coaching note:** a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

- **Visible cue:** the episode ends with the press down, a part in the gap, the part not in the tray or the bin, the gauge off the mat, an arm not home, or a gripper not fully open.
- **SOP rule broken:** Step 7. Check the press, the back stops, the part, the gauge, the bench, and the base. Then send both arms home with the grippers open and stop recording.
- **Coaching note:** check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see the die, the back stops, the lever, or the gauge on the part**, so nobody can tell if the sheet was against the stops, if the press cycled, or if a bend was good.
- **Hardware fault on an arm:** gripper failure, drift, a crash caused by the controller, or a motor error.
- **Base fault:** a lock that will not hold, or a base that rolls, slides, or turns with the lock on.
- **Faulty press brake mock:** a lever that sticks or binds, a punch that does not reach its top or bottom stop, a press set to the wrong angle, a back stop that moves on its own, or a press body that rocks on its bolts.
- **Faulty sheet:** one that arrives already bent, one with no lines or no numbers, one with lines drawn in the wrong place, or one that cracks at a bend made on its line against both stops.
- **Faulty gauge:** one with a bent or chipped edge, or one whose edges are not at 90 degrees.
- **Faulty tray:** pockets too small or too big for the part, or a tray that slides on the bench under a correct set down.
- **A place turns out to be too far for its arm** with the base parked correctly, so the top sheet, the front edge on the die, the lever knob, the gauge, the finished tray, or the reject bin cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Take the top sheet from the blank tray
2. Lay the sheet on the die
3. Push the sheet against the back stops
4. Pull the lever down and lift it up (bend 1)
5. Take the part out and give it a half turn
6. Lay the part on the die and push it against the back stops
7. Pull the lever down and lift it up (bend 2)
8. Take the part out and give it a half turn
9. Lay the part on the die and push it against the back stops
10. Pull the lever down and lift it up (bend 3)
11. Take the part out of the press
12. Lay the part on the gauge mat
13. Pick up the gauge
14. Gauge bend 1

15. Gauge bend 2
16. Gauge bend 3
17. Put the gauge back on the gauge mat
18. Put the part in the finished tray or the reject bin
19. Send both arms home and end the episode